NOTES ON NORMAL CROSSING DIVISORS

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Abstract. We establish basic facts surrounding the definition of normal crossing (nc) and simple normal crossing (snc) divisors on a smooth scheme over a field. We show that an effective divisor on a smooth scheme over a field is étale-locally the vanishing locus of a product of functions with independent differentials if, and only if, the same is true formally-locally; this is a property that characterizes normal crossing divisors. (To establish one of the implications we use the Artin approximation theorem, whose statement we recall, but do not prove.) We show that the same description holds Zariski-locally if, and only if, the divisor has normal crossings and smooth irreducible components, i.e., simple normal crossings. We also characterize snc divisors as reduced effective divisors such that, for every integer $r \geq 1$, the intersection of any r of their irreducible components is either smooth of codimension r or empty.

Let k be a field. Let X be a smooth k-scheme of pure dimension n. Let $D \subset X$ be an effective Cartier divisor.

Definition 1. Let r be an integer with $1 \leq r \leq n$. Let $\mathbf{A}^n$ denote the affine space over k with coordiantes $T_1, \ldots, T_n$. The *model normal crossings divisor* $Z_r \subset \mathbf{A}^r$ is the subscheme of defined by the vanishing of the product $T_1 \cdots T_r \in k[T_1, \ldots, T_r]$.

Definition 2. We say that D has *normal crossings* (*nc*, for short) at a point $x \in D$ if there exists a diagram (a "hat") of étale morphisms of pointed schemes

$$\begin{tikzcd} & (U, u) \arrow[dl, "g"'] \arrow[dr, "f"] & \\ (X, x) & & (\mathbf{A}^n, z) \end{tikzcd}$$

such that $g^{-1}D = f^{-1}Z_r$ for some r. We say that D has normal crossings if D has normal crossings at all of its points.

Definition 3. We say that D has *simple normal crossings* (*snc*, for short) at a point $x \in D$ if D has normal crossings at x and all of its (reduced) irreducible components are smooth in a Zariski neighborhood x. We say that D has *simple normal crossings* if D has simple normal crossings at all of its points.

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In the sequence we will make free use of the following facts:

- A map $h : T \to S$ of smooth schemes is étale if, and only if, its co-differential $dh^\vee : h^*\Omega_S \to \Omega_T$ is is an isomorphism.
- If T is a smooth k-scheme and $Z \subseteq T$ is a closed subscheme, then Z is smooth of codimension r in a Zariski neighborhood of a point $z \in T$ if, and only if, its ideal sheaf is is generated around z by r sections whose differentials have $k(z)$-linearly independent images in $\Omega_T \otimes k(z)$.

Example 4. Suppose that k has characteristic different from 2. Then the nodal cubic $C \subset \mathbf{A}^2$ with equation $y^2 = x^2(x+1)$ has normal crossings, but not simple normal crossings (since C is irreducible, but not smooth).

To see that C has normal crossings at the origin, suppose that $g : (U, u) \to (\mathbf{A}^2, 0)$ is an étale morphism of pointed schemes such that $x + 1 = z^2$ for some unit $z \in \Gamma(U, \mathscr{O}_U)^\times$. We will show that such étale morphisms exist in a moment. Granting this, the inverse image $g^{-1}D \subset U$ has equation

$$(y + xz)(y - xz) = 0.$$

Because $f : U \to \mathbf{A}^2$ is étale, the differentials dx and dy form a basis for Ω_U as an $\mathscr{O}_U$-module. From the equation $z^2 = x + 1$, it follows that dz satisfies the relation $2z\ dz = dx$, so that

$$d(xz) = x\ dz + z\ dx = \left(\frac{x}{2z} + z\right) dx.$$

Because x vanishes in $k(u)$, we may assume after shrinking U to a Zariski neighborhood of u that $d(xz)$ and dy form a basis for Ω_U. Then the map of pointed schemes

$$f := (y + xz, y - xz) : (U, u) \to (\mathbf{A}^2, 0)$$

is étale and satisfies $g^{-1}D = f^{-1}Z_2$.

Let us now find an étale morphism $g : (U, u) \to (\mathbf{A}^2, 0)$ as above. Let $g : Y \to \mathbf{A}^2$ be the double cover that branches over the line $x + 1 = 0$. Thus Y is the subscheme of the affine space $\mathbf{A}^3$ with coordinates x, y, z defined by the equation $z^2 = x + 1$. The differential of this equation is $2z\ dz - dx$, so Y is smooth and

$$\Omega_Y = (\Omega_{\mathbf{A}^3}/\langle 2z\ dz - dx\rangle)|_Y$$

has basis dy and dz. The covering map $g : Y \to \mathbf{A}^2$ pulls back $dx \mapsto 2z\ dz$ and $dy \mapsto dy$, so is étale over the open subset $U \subset Y$ where $z \neq 0$. Note that $g(U) \subset \mathbf{A}^2$ is an open subset where $x + 1 \neq 0$, and in particular contains the origin. This completes the proof that C has normal crossings at that point.

It is easy to check that C is smooth away from the origin, so using Corollary 7 below we conclude that C has normal crossings everywhere.

To see that C is irreducible, note that the inverse image $g^{-1}C \subseteq Y$ is the union of the subschemes of Y defined by $y + xz = 0$ and $y - xz = 0$. Inside $\mathbf{A}^3$, these subschemes are defined by $x = z^2 - 1$ and $y = \mp xz$, so each is isomorphic to $\mathbf{A}^1$ with coordinate z. Thus the inverse image $g^{-1}D \subset Y$

consists of two irreducible components. It is easy to check that, if $\bar{k}$ is an algebraic closure of k, then the set of $\bar{k}$-points of either of these components surjects onto the the set of $\bar{k}$-points of C. Hence C is irreducible, as claimed.

Proposition 5.

(1) *The set of points at which D has normal crossings is open.*
(2) *The divisor D has normal crossings if, and only if, it has normal crossings at all of its closed points.*
(3) *If D has normal crossings, then D is reduced.*

Proof. The diagram that displays D as having normal crossings at a point $x \in X$ does so for all points of the image $g(U) \subset X$. Thus (1) follows from the fact that étale morphisms are open. Clearly, (1) implies (2). If $h : T \to S$ is an étale morphism of schemes, then T is reduced if, and only if, $h(T)$ is reduced. Thus (3) follows from the observation that the model normal crossing divisors are reduced. □

Proposition 6. *Let $x \in D$ be a point. The divisor D has normal crossings at x if, and only if, there exists an étale morphism of pointed schemes $g : (U, u) \to (X, x)$ and sections $f_1, \ldots, f_r \in \Gamma(U, \mathscr{O}_U)$ such that*

(1) *$g^{-1}D$ equals the closed subscheme $\{f_1 \cdots f_r = 0\} \subseteq U$, and*
(2) *$df_1, \ldots, df_r$ have $k(u)$-independent images in $\Omega_U \otimes k(u)$.*

Proof. Suppose that D has normal crossings at x. Choose étale maps $g : (U, u) \to (X, x)$ and $f : (U, u) \to (\mathbf{A}^n, z)$ that exhibit the normal crossings. For $i = r + 1, \ldots, n$, let $f_i := f^{\#}T_i$, so that $df_i = f^*dT_i$ for all $i = 1, \ldots, n$. Then (1) and (2) hold.

Suppose conversely that there exist $g : (U, u) \to (X, x)$ and $f_1, \ldots, f_r \in \Gamma(U, \mathscr{O}_U)$ such that (1) and (2) hold. Let $f_{r+1}, \ldots, f_n \in \mathscr{O}_{U,u}$ be elements whose differentials complete the images of $df_1, \ldots, df_r$ to a $k(u)$-basis of $\Omega_U \otimes k(u)$. These elements exist because Ω_U is generated as an $\mathscr{O}_U$-module by the image of the universal derivation $d : \mathscr{O}_U \to \Omega_U$. After shrinking U to a Zariski neighborhood of u, we may assume that $f_{r+1}, \ldots, f_n \in \Gamma(U, \mathscr{O}_U)$. By Nakayama's lemma, after shrinking U further, we may also assume that $df_1, \ldots, df_n$ generate and therefore provide a basis for Ω_U as an $\mathscr{O}_U$-module. This implies that map

$$f := (f_1, \ldots, f_n) : U \to \mathbf{A}^n$$

induces an isomorphism $f^*\Omega_{\mathbf{A}^n} \to \Omega_X$, hence is étale. The étale maps f and g exhibit D as having normal crossings at x. □

Corollary 7. *Let $x \in D$ be a point. If D is smooth at x, then D has normal crossings at x.*

Proof. The differential of any (Zariski-)local equation for D near x has nonzero image in $\Omega_X \otimes k(x)$, so the result follows from Proposition 6. □

Proposition 8. *Let $x \in D$ be a closed point. Suppose that the finite field extension $k \subseteq k(x)$ is separable. The divisor D has normal crossings at x if, and only if, there exists an étale morphism of pointed schemes $g : (U, u) \to (X, x)$ and a regular system of parameters $f_1, \ldots, f_n \in \mathscr{O}_{U,u}$ such that $f_1 \cdots f_r \in \mathscr{O}_{U,u}$ is a local equation for $g^{-1}D \subseteq U$.*

The requirement that the finite field extension $k \subseteq k(x)$ be separable is satisfied for example when k is algebraically closed or has characteristic zero.

Proof. If $g : (U, u) \to (X, x)$ is an étale morphism of pointed schemes, then the field extension $k(x) \subseteq k(u)$ is finite separable. It follows that $k \subseteq k(u)$ is finite separable, hence that the natural map

$$\mathfrak{m}_u/\mathfrak{m}_u^2 \to \Omega_U \otimes k(u)$$

sending $\bar{g} \mapsto dg$ is an isomorphism. Elements $f_1, \ldots, f_n \in \mathfrak{m}_u$ form a regular system of parameters for $\mathscr{O}_{U,u}$ if, and only if, their images form a $k(u)$-basis for $\mathfrak{m}_u/\mathfrak{m}_u^2$. Therefore the result follows from Proposition 6. □

Lemma 9. *Let $(A, \mathfrak{m}, \kappa)$ be a local k-algebra. Let $f \in \mathfrak{m}$ be an element such that df has nonzero image in $\Omega_A \otimes \kappa$. Then f is irreducible.*

Proof. Suppose that $f = gh$. By assumption

$$df = hdg + gdh$$

has nonzero image in $\Omega_A \otimes \kappa$. Therefore h and g cannot both map to zero in κ, so one of them must be a unit in A. □

Proposition 10. *If D has normal crossings, then each of its irreducible components has normal crossings.*

Proof. Let $x \in D$ be a point. Let $V \subset X$ be a quasi-compact Zariski neighborhood of x. (If X is of finite type, we may take $V = X$.) Let $D_1, \ldots, D_m$ be the irreducible components of $D \cap V$. Let $g : (U, u) \to (V, x)$ be an étale morphism of pointed schemes and $f_1, \ldots, f_r \in \Gamma(U, \mathscr{O}_U)$ sections with independent differentials at x such that the subscheme $g^{-1}(D \cap V) \subseteq U$ defined by $f_1 \cdots f_r = 0$. For each $j = 1, \ldots, m$, let $h_j \in \mathscr{O}_{U,u}$ be a local equation for $g^{-1}D_i$. Then $(f_1 \cdots f_r) = (h_1 \cdots h_m)$ as ideals of $\mathscr{O}_{U,u}$. The nonunits among $f_1, \ldots, f_r \in \mathscr{O}_{U,u}$ are irreducible as their differentials have nonzero images in $\Omega_U \otimes k(u)$. Because U is smooth over k, the local ring $\mathscr{O}_{U,u}$ is regular and hence a UFD. It follows that each h_j can be expressed in $\mathscr{O}_{U,u}$ as a unit times the product of a subset of $\{f_1, \ldots, f_r\}$. □

Proposition 11. *Let $x \in D$ be a point. The following statements are equivalent:*

1. *D has simple normal crossings at x.*
2. *D is reduced and, for every integer $c \geq 1$, the scheme theoretic intersection of any c (reduced) irreducible components of D that contain x is smooth of codimension c in Zariski neighborhood of x.*

(3) *There exists a Zariski neighborhood* $V \subset X$ *of* x *and sections* $f_1, \ldots, f_r \in \Gamma(V, \mathscr{O}_V)$ *such that*
   (a) $D \cap U = \{f_1 \cdots f_r = 0\}$ *as subschemes of* U*; and*
   (b) *the differentials* $df_1, \ldots, df_r$ *have* $k(x)$*-linearly independent images in* $\Omega_X \otimes k(x)$*.*

Proof. The three statements being Zariski-local around $x \in X$, we may assume that X is quasi-compact. Then D has finitely many irreducible components. Let $D_1, \ldots, D_m$ denote the components that contain x.

Let us show that (3) implies (1). If (3) holds, then clearly D has normal crossings at x. Let $f_1, \ldots, f_r \in \Gamma(V, \mathscr{O}_V)$ be as in (3). Remove from $f_1, \ldots, f_r$ the sections that are units in $\mathscr{O}_{X,x}$ and shrink V around x so as to make sure that (3a) continues to hold. For each $i = 1, \ldots, r$, the divisor $D'_i \subset V$ defined by the vanishing of f_i is irreducible and smooth in a Zariski neighborhood of x, as df_i has nonzero image in $\Omega_X \otimes k(x)$. Shrinking V, we may assume each D'_i is irreducible and smooth. If $D'_i = D'_j$, then $f_i = uf_j$ for some unit $u \in \mathscr{O}_{X,x}$, so that

$$df_i(x) = u(x)df_j(x) + f_j(x)du(x) = u(x)df_j(x)$$

in $\Omega_X \otimes k(x)$. By (3b) this is only possible if $i = j$. Hence the divisors $D'_1, \ldots, D'_r \subset V$ are distinct. It follows that $r = m$ and that $D_1 \cap V, \ldots, D_m \cap V$ are $D'_1, \ldots, D'_m$ in some order. In particular, $D_1 \cap V, \ldots, D_m \cap V$ are smooth, so (1) holds.

Assume (1). Let us show that (2) holds. Let $g : (U, u) \to (X, x)$ be an étale morphism of pointed schemes and $f_1, \ldots, f_r \in \Gamma(U, \mathscr{O}_U)$ be sections such that $g^{-1}D \subseteq U$ is the subscheme $f_1 \cdots f_r = 0$ and such that $df_1, \ldots, df_r$ are linearly independent in $\Omega_U \otimes k(u)$. By assumption, the divisors $D_1, \ldots, D_m$ are smooth at x, so their inverse images in U are smooth at u. After shrinking U to a Zariski neighborhood of u, we may assume that $g^{-1}D_1, \ldots, g^{-1}D_m$ are the irreducible components of $g^{-1}D$. By the discussion of the preceding paragraph, after shrinking U further and reordering $f_1, \ldots, f_r$, we may assume that $m = r$ and that, for each $i = 1, \ldots, m$, the subscheme $g^{-1}D_i \subset U$ is defined by the vanishing of $f_i \in \Gamma(U, \mathscr{O}_U)$. Let c be an integer such that $1 \leq c \leq m$ and consider the scheme theoretic intersection of c irreducible components among $D_1, \ldots, D_m$. After reordering $D_1, \ldots, D_m$, we may assume that these are $D_1, \ldots, D_c$. To show that $D_1 \cap \cdots \cap D_c$ is smooth of codimension c at x, it suffices to show that the same is true for the inverse image $g^{-1}(D_1 \cap \cdots \cap D_c)$ at u. Now, the ideal of this closed subscheme of U is generated by the sections $f_1, \ldots, f_c \in \Gamma(U, \mathscr{O}_U)$, whose differentials are independent in $\Omega_U \otimes k(u)$ by hypothesis. This independence implies that $g^{-1}(D_1 \cap \cdots \cap D_c)$ is smooth of codimension c around u. Thus (2) holds.

Suppose that (2) holds. Let us show (3). Choose $V \subset X$ a Zariski neighborhood of x and $f_1, \ldots, f_m \in \Gamma(V, \mathscr{O}_V)$ be sections such that $D_i \cap V$ is the closed subscheme of V defined by the vanishing of f_i for all $i =$

$1, \ldots, m$. Let $I \subseteq \mathscr{O}_X$ be the ideal of the scheme theoretic intersection $Y := D_1 \cap \cdots \cap D_m$. Then I is generated by $f_1, \ldots, f_m$. By assumption, Y is smooth of codimension m in a neighborhood of x in $V \subseteq X$. This implies that $df_1, \ldots, df_m$ have $k(x)$-linearly independent images in $\Omega_X \otimes k(x)$, i.e., that (3b) is satisfied. It also shows that the divisor $\{f_1 \cdots f_r = 0\} \subseteq V$ has normal crossings at x, so by shrinking V around x, we may assume this divisor is reduced. (To see that $f_1 \cdots f_r = 0$

defines a reduced divisor around $x \in V$, one could also argue directly using the fact that $\mathscr{O}_{X,x}$, which is a UFD.) By assumption, $D \subset X$ is also reduced, and by construction $\{f_1 \cdots f_r = 0\}$ and $D \cap V$ have the same support. Therefore the ideal of $D \cap V$ is generated by $f_1 \cdots f_m$, so (3a) is also satisfied. □

Corollary 12. *Let $D_1, \ldots, D_m$ denote the irreducible components of D. The divisor D has simple normal crossings if, and only if, the following conditions are satisfied:*

1. *D is reduced;*
2. *D_1 is smooth;*
3. *$\sum_{i=2}^m D_i$ has snc; and*
4. *D_1 meets the intersection of any number of the divisors $D_2, \ldots, D_m$ transversely.*

Corollary 13. *Let $x \in D$ be a point. The divisor D has normal crossings at x if, and only if, there exists an étale morphism of pointed schemes $g : (U, u) \to (X, x)$ such that $g^{-1}D \subseteq U$ has simple normal crossings at u.*

We now recall the statement of the Artin approximation theorem.

Definition 14. Let S be a scheme. An *S-algebra* is a ring B equipped with a morphism $\operatorname{Spec} B \to S$. A morphism of S-algebras is a ring homomorphism $\phi : B \to B'$ such that $\operatorname{Spec}(\phi) : \operatorname{Spec} B' \to \operatorname{Spec} B$ is a morphism of S-schemes.

Definition 15. Let S be a scheme. A contravariant functor

$$F : (\mathrm{Sch}/S) \to (\mathrm{Set})$$

is *limit preserving* if for every direct limit of S-algebras $\varinjlim B_\lambda$ the natural map

$$\varinjlim F(B_\lambda) \to F(\varinjlim B_\lambda)$$

is bijective. (Here and in what follows, we write $F(B)$ instead of $F(\operatorname{Spec} B)$ when B is an S-algebra.)

Theorem 16 (Artin approximation)**.** *Let S be a scheme locally of finite type over k.*

$$F : (\mathrm{Sch}/S) \to (\mathrm{Set})$$

be a limit preserving contravariant functor. Let $s \in S$ be a closed point. Let $\widehat{\xi} \in F(\widehat{\mathscr{O}}_{S,s})$. For every integer $N \geq 1$, there exists a finite type morphism of pointed k-schemes $(U, u) \to (S, s)$ and an element $\xi \in F(U)$ such that

(1) *the natural map* $\widehat{\mathscr{O}}_{U,u} \to \widehat{\mathscr{O}}_{S,s}$ *is an isomorphism, and*
(2) *the restrictions of* $\widehat{\xi}$ *and* ξ *to* $F(\mathscr{O}_{U,u}/\mathfrak{m}_u^N)$ *are equal (under the identifications* $\mathscr{O}_{U,u}/\mathfrak{m}_u^N \cong \mathscr{O}_{S,s}/\mathfrak{m}_s^N \cong \widehat{\mathscr{O}}_{S,s}/\widehat{\mathfrak{m}}_s^N$*).*

By EGA, Proposition IV.17.6.3, a finite type morphism $(U,u) \to (S,s)$ satisfies condition condition (1) if, and only if, it is étale in a Zariski neighborhood of u and induces an isomorphism of residue fields $k(s) \xrightarrow{\sim} k(u)$.

Theorem 16 is a special case of the result proved in the Stacks Project, Tag 07XB.

Proposition 17. *Let* $x \in D$ *be a closed point such that the finite extension* $k \subseteq k(x)$ *is separable, and such that the residue field* $k(x)$ *is separably closed. The divisor* D *has normal crossings at* x *if, and only if, there exists a regular system of parameters* $f_1, \dots, f_n \in \widehat{\mathscr{O}}_{X,x}$ *such that the ideal of the closed subscheme*

$$D \times_X \operatorname{Spec} \widehat{\mathscr{O}}_{X,x} \subseteq \operatorname{Spec} \widehat{\mathscr{O}}_{X,x}$$

is generated by the product $f_1 \cdots f_r \in \widehat{\mathscr{O}}_{X,x}$ *for some* r*.*

Note that the assumptions that $k \subset k(x)$ is be separable and $k(x)$ is separably closed are satisfied if k has either characteristic zero or is algebraically closed.

Proof. Suppose that D has normal crossings at x. Because $k \subset k(x)$ is separable, there exists an étale morphism of pointed k-schemes $g : (U,u) \to (X,x)$ and a regular system of parameters $f_1, \dots, f_n \in \mathscr{O}_{U,u}$ such that the ideal of $D \times_X \operatorname{Spec} \mathscr{O}_{U,u}$ is generated by $f_1 \cdots f_r$. Because $k(x)$ is separably closed and $g : U \to X$ is étale, the natural map $\widehat{\mathscr{O}}_{X,x} \to \widehat{\mathscr{O}}_{U,u}$ is an isomorphism (EGA, Proposition IV.17.6.3). Therefore $f_1, \dots, f_n$ induce the desired regular system of parameters on $\widehat{\mathscr{O}}_{X,x}$.

Suppose conversely that $f_1, \dots, f_n \in \widehat{\mathscr{O}}_{X,x}$ is a regular system of parameters such that $f_1 \cdots f_r = 0$ is an equation for $D \times_X \operatorname{Spec} \widehat{\mathscr{O}}_{X,x}$. To show that D has normal crossings at x, we may replace X with any Zariski neighborhood of x. Hence we may assume that $D \subseteq X$ is defined by the vanishing of a global section $\phi \in \Gamma(X, \mathscr{O}_X)$.

Let

$$Z_r \subset \mathbf{A}^n = \operatorname{Spec} k[T_1, \dots, T_n]$$

denote the subscheme defined by $T_1 \cdots T_r = 0$. Let

$$F : (\mathrm{Sch}/X) \to (\mathrm{Set})$$

be the contravariant functor that sends a scheme T over X to the set of k-morphisms $h : T \to \mathbf{A}^n$ such that $h^{-1} Z_r = D \times_X T$. Given a morphism of schemes $\operatorname{Spec} B \to X$, we have

$$F(B) = \{(h_1, \dots, h_n) \in B^{\oplus r} \mid h_1 \dots h_r = u\phi \text{ for some unit } u \in B^\times\}.$$

Thus F is limit preserving.

We are given an element $f = (f_1, \ldots, f_r) \in F(\widehat{\mathscr{O}}_{X,x})$. By Artin approximation (Theorem 16), we may find an étale morphism of pointed schemes $g : (U, u) \to (X, x)$ and an element $h = (h_1, \ldots, h_n) \in F(U)$ such that the restrictions of f and h to $\mathscr{O}_{U,u}/\mathfrak{m}_u^2$ are equal. Then $h_1, \ldots, h_r \in \mathscr{O}_{U,u}$ are a regular system of parameters: these elements lie in $\mathfrak{m}_u$ and have linearly independent images in $\mathfrak{m}_u/\mathfrak{m}_u^2 \cong \Omega_U \otimes k(u)$. Shrinking U to a Zariski neighborhood of u, we may assume that $dh_1, \ldots, dh_r \in \Gamma(U, \Omega_U)$ generate and therefore form a basis for Ω_U. Then $g : (U, u) \to (X, x)$ and $h : (U, u) \to (\mathbf{A}^r, h(u))$ exhibit D having normal crossings at x. $\square$